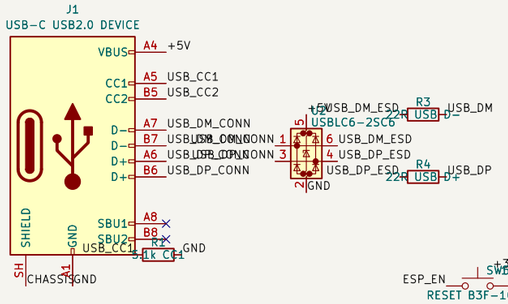
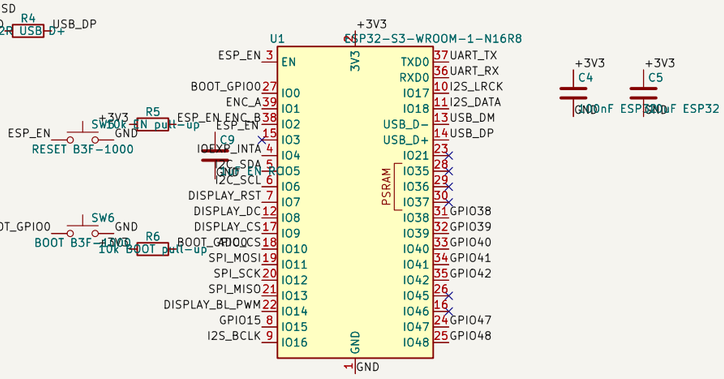
ESP-DubSiren V1 — ESP32-S3 native USB / PCM5122 / MCP3208 / MCP23017

USB-C DEVICE + ESD + 3V3 POWER

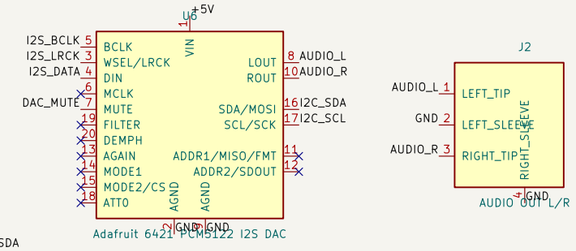


ESP32-S3-WROOM-1-N16R8

PCM5102A BREAKOUT FOOTPRINT IS PROVISIONAL — MEASURE PHYSICAL MODULE BEFORE FABRICATION



DIGITAL AUDIO / REAR OUTPUT

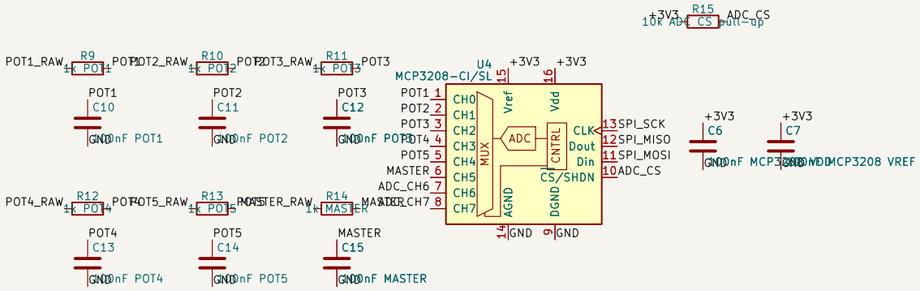


I/O EXPANDER + FRONT CONTROLS

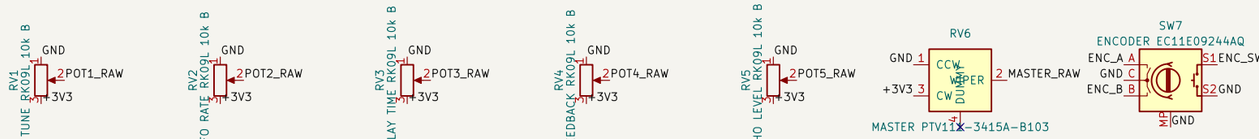
DISPLAY / EXPANSION HEADERS

V0 MECHANICAL PROTOTYPE: CONTROL NETS ARE INTENTIONALLY NOT ASSIGNED TO ESP32 GPIO

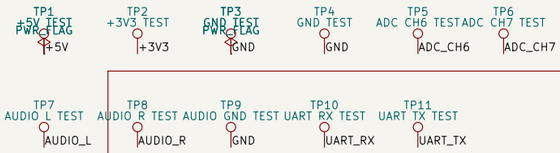
ANALOG CONTROLS + 1k/100nF INPUT FILTERS







TEST POINTS



DESIGN NOTE: master uses PTV111-3415A-B103 (rear-facing shaft). The originally requested -1415A is vertical-shaft and mechanically incompatible with a rear panel.

CHASSIS is isolated from circuit GND by default: R17/C16 are DNP configuration options. USB shield and connector shell land on CHASSIS.

PCM5122 module U6 follows Adafruit 6421 dual-header pin order; MCLK is intentionally unused. Native USB D+/D- include USBLC6-2 and 22R series damping.

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